

Some Basic Concept of Chemistry

Unit 1: Some Basic Concepts of Chemistry

1. What is Science and Chemistry?

- Science is a continuing human effort to systematize knowledge for describing and understanding nature.
- It is sub-divided into various disciplines like chemistry, physics, biology, and geology.
- Chemistry is the branch of science that studies the preparation, properties, structure, and reactions of material substances.
- It is also called the science of molecules and their transformations. It focuses not just on the 100 elements, but on the infinite variety of molecules that can be built from them.
- Chemistry deals with the composition, structure, properties, and interaction of matter. These aspects are best understood in terms of basic constituents: atoms and molecules.

2. Development of Chemistry

- Chemistry, as understood today, is not a very old discipline.
- Historically, it developed from the search for:
 - Philosopher's stone (Paras), which was believed to convert base metals (like iron and copper) into gold.
 - 'Elixir of life', which would grant immortality.
- People in ancient India had significant knowledge of scientific phenomena before modern science.
- Chemistry mainly developed in the form of Alchemy and Iatrochemistry during 1300-1600 CE.
- Modern chemistry took shape in the 18th century Europe, influenced by alchemical traditions introduced by Arabs.
- Other cultures, including Chinese and Indian, had their own alchemical traditions with extensive knowledge of chemical processes and techniques.

3. Chemistry in Ancient India

- In ancient India, chemistry was known as Rasayan Shastra, Rastantra, Ras Kriya, or Rasvidya.

- It encompassed:

- o Metallurgy

- o Medicine

- o Manufacture of cosmetics, glass, dyes, etc.

- Archaeological evidence from Mohenjodaro (Sindh) and Harappa (Punjab) indicates a long history of chemistry development in India.

- o Baked bricks were used in construction, indicating mass production of pottery, an early chemical process involving mixing, molding, and heating materials.

- o Glazed pottery found in Mohenjodaro.

- o Gypsum cement (containing lime, sand, and CaCO_3) used in construction.

- o Harappans made faience, a type of glass for ornaments.

- o They melted and forged various objects from lead, silver, gold, and copper.

- o Improved copper hardness by using tin and arsenic for artifacts.

- o Glass objects found in Maski (South India, 1000–900 BCE) and Hastinapur and Taxila (North India, 1000–200 BCE).

- o Glass and glazes were colored using metal oxides.

- Copper metallurgy dates back to the beginning of chalcolithic cultures in India, with indigenous development of extraction technologies for copper and iron.

- Rigveda mentions tanning of leather and dyeing of cotton (1000–400 BCE).

- The golden gloss of black polished ware of northern India remains a "chemical mystery" and shows mastery of kiln temperature control.

- Kautilya's Arthashastra describes salt production from the

sea.

- Ancient Vedic literature statements often agree with modern scientific findings.
- Archaeological sites in north India have yielded copper utensils, iron, gold, silver ornaments, terracotta discs, and painted grey pottery.
- Sushruta Samhita explains the importance of Alkalies.
- Charaka Samhita indicates ancient Indians knew how to prepare:
 - o Sulphuric acid, nitric acid
 - o Oxides of copper, tin, and zinc
 - o Sulphates of copper, zinc, and iron
 - o Carbonates of lead and iron.
 - o It also discusses the reduction of particle size of metals (nanotechnology) and the use of bhasma (nanoparticles of metals) in treating ailments.
- Rasopanishada describes gunpowder mixture preparation.
- Tamil texts detail fireworks preparation using sulfur, charcoal, saltpetre (potassium nitrate), mercury, camphor, etc..
- Nagarjuna (a great Indian scientist, chemist, alchemist, and metallurgist) wrote Rasratnakar, dealing with mercury compounds and metal extraction methods (gold, silver, tin, copper).
- Rsarnavam (around 800 CE) discusses various furnaces, ovens, crucibles, and methods for identifying metals by flame color.
- Chakrapani discovered mercury sulfide and is credited with inventing soap using mustard oil and alkalies; Indians began making soaps in the 18th century CE, also using Eranda oil, Mahua seeds, and calcium carbonate.
- Ajanta and Ellora paintings show a high level of ancient Indian science due to their preserved freshness.

- Varāhmihir's Brihat Samhita (6th century CE encyclopaedia) describes preparing a glutinous material for walls and roofs from plant extracts, fruits, seeds, barks, and resins.
 - Atharvaveda (1000 BCE) mentions dye stuffs like turmeric, madder, sunflower, orpiment, cochineal, lac, kamplcica, pattanga, and jatuka.
 - Varāhmihir's Brihat Samhita also gives recipes for perfumes and cosmetics, including hair dyeing with indigo, iron powder, black iron/steel, and acidic sour rice gruel extracts.
 - Gandhayukli describes recipes for scents, mouth perfumes, bath powders, incense, and talcum powder.
 - Paper was known in India by the 17th century (Chinese traveler I-tsing's account).
 - Ink was used in India from the 4th century (Taxila excavations), with colors from chalk, red lead, and minimum.
 - Fermentation was well-known to Indians, with Vedas and Kautilya's Arthashastra mentioning many types of liquors. Charaka Samhita also lists ingredients for Asavas (barks, stems, flowers, leaves, woods, cereals, fruits, sugarcane).
 - The concept of matter being made of indivisible building blocks (atomic theory) appeared in India centuries BCE.
- Acharya Kanda (born 600 BCE), originally Kashyap, was the first proponent of this theory, naming these small, indivisible particles 'Paramānu' (comparable to atoms).
- o He authored Vaiseshika Sutras.
 - o Paramānu are eternal, indestructible, spherical, suprasensible, and in motion.
 - o They cannot be sensed by human organs.
 - o There are varieties of atoms, and they can form pairs or triplets through unseen forces.
 - o Kanda conceptualized this theory around 2500 years before John Dalton.

- After the decline of alchemy, iatrochemistry declined with the introduction of Western medicine in the 20th century. Indigenous techniques declined as foreign products entered the market. Modern science appeared in India in the later 19th century.

4. Importance and Role of Chemistry Today

- Chemistry plays a central role in science and is intertwined with other branches.
- Its principles apply to:
 - o Weather patterns
 - o Brain functioning
 - o Computer operations
 - o Chemical industries (fertilizers, alkalis, acids, salts, dyes, polymers, drugs, soaps, detergents, metals, alloys, new materials).
- National Economy: Contributes significantly to the economy and generates employment by establishing industries.
- Human Needs: Helps meet needs for food, healthcare products, and improved quality of life.
 - o Large-scale production of fertilizers, improved pesticides, and insecticides.
 - o Methods for isolating and synthesizing life-saving drugs (e.g., cisplatin, taxol for cancer; AZT for AIDS).
- Development and Growth of a Nation: Better understanding of chemical principles allows for designing and synthesizing new materials with specific magnetic, electric, and optical properties (e.g., superconducting ceramics, conducting polymers, optical fibers).
- Environmental Degradation: Helped in dealing with problems like ozone depletion by synthesizing safer alternatives to CFCs.
- Future Challenges: Management of greenhouse gases (methane, carbon dioxide), understanding biochemical

processes, using enzymes for large-scale chemical production, and synthesizing new exotic materials are intellectual challenges for future chemists. India needs talented and creative chemists for these challenges.

5. Nature of Matter

- Matter: Anything that has mass and occupies space (e.g., book, pen, water, air, living beings).
- States of Matter: Matter exists in three physical states:
 - o Solid:
 - Particles are held very close in an orderly fashion with little freedom of movement.
 - Have definite volume and definite shape.
 - o Liquid:
 - Particles are close but can move around.
 - Have definite volume but no definite shape (take container's shape).
 - o Gas:
 - Particles are far apart with easy and fast movement.
 - Have neither definite volume nor definite shape (completely occupy container space).
- Interconversion of States: States are interconvertible by changing temperature and pressure (e.g., solid $\rightarrow$ liquid $\rightarrow$ gas on heating; gas $\rightarrow$ liquid $\rightarrow$ solid on cooling).

6. Classification of Matter

- At the macroscopic level, matter is classified as mixture or pure substance.
- Pure Substance:
 - o All constituent particles are the same in chemical nature.
 - o Have fixed composition (e.g., copper, silver, gold, water, glucose).
 - o Constituents cannot be separated by simple physical methods.

- o Further classified into elements and compounds.

- Mixture:

- o Contains two or more pure substances that can be present in any ratio.

- o Composition is variable (e.g., sugar solution, air, tea).

- o Components can be separated by physical methods (hand-picking, filtration, crystallization, distillation).

- o Homogeneous Mixture: Components completely mix and are uniformly distributed (e.g., sugar solution, air).

- o Heterogeneous Mixture: Composition is not uniform, and components can sometimes be visible (e.g., salt and sugar mixture, grains with dirt).

- Elements:

- o Particles consist of only one type of atoms.

- o Particles may exist as atoms or molecules.

- o All atoms of one element are of one type, but atoms of different elements differ in nature.

- o Examples: Sodium (Na) or copper (Cu) have atoms as constituent particles. Hydrogen (H_2), nitrogen (N_2), and oxygen (O_2) gases consist of molecules formed by two or more atoms.

- Compounds:

- o Formed when two or more atoms of different elements combine in a definite ratio.

- o Properties of a compound are different from those of its constituent elements (e.g., hydrogen and oxygen are gases, but water (H_2O) is a liquid; hydrogen burns, oxygen supports combustion, but water is a fire extinguisher).

- o Constituents cannot be separated by physical methods; can be separated by chemical methods.

- o Examples: Water (H_2O - two hydrogen, one oxygen), carbon dioxide (CO_2 - one carbon, two oxygen).

7. Properties of Matter and Their Measurement

- Every substance has unique or characteristic properties.
- Physical Properties: Can be measured or observed without changing the identity or composition of the substance.
 - Examples: Color, odor, melting point, boiling point, density.
- Chemical Properties: Measurement or observation requires a chemical change to occur.
 - Examples: Composition, combustibility, reactivity with acids and bases, acidity, basicity.
- Chemists describe, interpret, and predict substance behavior based on knowledge of their physical and chemical properties, determined by careful measurement and experimentation.
- Quantitative Measurement: Represented by a number followed by units (e.g., 6 m).
- Systems of Measurement:
 - Historically, English System and Metric System were used.
 - Metric system (from France, late 18th century) was more convenient due to its decimal basis.
 - International System of Units (SI): Established in 1960 by the 11th General Conference on Weights and Measures (CGPM) to create a common standard system.
 - Based on seven base units pertaining to fundamental scientific quantities:
 1. Length: metre (m)
 2. Mass: kilogram (kg)
 3. Time: second (s)
 4. Electric current: ampere (A)
 5. Thermodynamic temperature: kelvin (K)
 6. Amount of substance: mole (mol)
 7. Luminous intensity: candela (cd)
 - Other physical quantities (speed, volume, density) are derived from these base units.

□ SI allows prefixes to indicate multiples or submultiples of a unit (e.g., kilo, milli, micro, nano).

□ National Metrology Institutes (NMIs), like India's National Physical Laboratory (NPL) in New Delhi, maintain measurement standards and realize base and derived units.

□ Reference Standards:

□ Kilogram: Since 1889, defined as the mass of a platinum-iridium (Pt-Ir) cylinder stored in Sevres, France, chosen for its resistance to chemical attack. Scientists are researching new standards, possibly using Avogadro constant determination.

□ Metre: Originally defined by marks on a Pt-Ir bar (0°C); redefined in 1960 by krypton laser wavelength; redefined in 1983 by CGPM as the length of path travelled by light in vacuum during $1/299,792,458$ of a second.

8. Common Physical Quantities and Their Measurement

• Mass vs. Weight:

o Mass: The amount of matter present in a substance; constant.

o Weight: The force exerted by gravity on an object; varies with gravity.

o Mass is accurately determined using an analytical balance.

o SI unit of mass is kilogram (kg); gram (g) is often used in labs ($1\text{ kg} = 1000\text{ g}$).

• Volume:

o The amount of space occupied by a substance.

o SI unit: m^3 (length^3).

o Commonly used units in labs: cm^3 or dm^3 .

o Litre (L) is a common non-SI unit for liquids: $1\text{ L} = 1000\text{ mL} = 1000\text{ cm}^3 = 1\text{ dm}^3$.

o Measured by graduated cylinders, burettes, pipettes; volumetric flasks prepare known volumes.

• Density:

- o Mass per unit volume: Density = Mass / Volume.
- o SI unit: kg m^{-3} .
- o Commonly expressed in labs as g cm^{-3} .
- o Higher density means particles are more closely packed.
- Temperature:
 - o Common scales: $^{\circ}\text{C}$ (Celsius), $^{\circ}\text{F}$ (Fahrenheit), K (Kelvin).
 - o Kelvin (K) is the SI unit.
 - o Celsius calibrated 0°C (freezing point of water) to 100°C (boiling point of water). Fahrenheit is 32°F to 212°F .
 - o Relationships:
 - $\square \text{ } ^{\circ}\text{F} = (9/5)^{\circ}\text{C} + 32$
 - $\square \text{ K} = ^{\circ}\text{C} + 273.15$
 - o Negative temperatures are possible in Celsius, but not in Kelvin.

9. Uncertainty in Measurement

- Every experimental measurement has some uncertainty due to instrument limitations and human skill.
- Scientific Notation:
 - o Used to conveniently represent extremely large or small numbers encountered in chemistry (e.g., number of molecules, mass of an atom).
 - o Form: $N \times 10^n$, where N (digit term) is between 1.000... and 9.999..., and n is a positive or negative exponent.
 - o Example: $232.508 = 2.32508 \times 10^2$; $0.00016 = 1.6 \times 10^{-4}$.
 - o Rules for mathematical operations (multiplication, division, addition, subtraction) follow exponential rules.
- Significant Figures:
 - o Meaningful digits known with certainty, plus one estimated/uncertain digit.
 - o Indicate the uncertainty in experimental or calculated values.
 - o Uncertainty of ± 1 in the last digit is generally understood.
 - o Rules for determining significant figures:

1. All non-zero digits are significant (e.g., 285 cm has three; 0.25 mL has two).
2. Zeros preceding the first non-zero digit are NOT significant (e.g., 0.03 has one; 0.0052 has two).
3. Zeros between two non-zero digits ARE significant (e.g., 2.005 has four).
4. Zeros at the end or right of a number ARE significant if they are to the right of the decimal point (e.g., 0.200 g has three).
5. Terminal zeros are NOT significant if there is no decimal point (e.g., 100 has one).
6. Numbers like 100 are better represented in scientific notation to show significant figures (e.g., 1×10^2 for one, 1.0×10^2 for two, 1.00×10^2 for three).
7. In scientific notation, all digits are significant (e.g., 4.01×10^2 has three; 8.256×10^{-3} has four).
8. Counting numbers of objects (e.g., 2 balls, 20 eggs) have infinite significant figures as they are exact numbers.

o Operations with Significant Figures:

- Addition and Subtraction: The result cannot have more digits to the right of the decimal point than the original number with the fewest digits after the decimal point.
- Multiplication and Division: The result must be reported with no more significant figures than the measurement with the fewest significant figures.

o Rounding Off Numbers:

0. If the rightmost digit to be removed is more than 5, increase the preceding number by one (e.g., 1.386 rounds to 1.39).
1. If the rightmost digit to be removed is less than 5, the preceding number is not changed (e.g., 4.334 rounds to 4.33).
2. If the rightmost digit to be removed is 5:

□ Preceding number is not changed if it is an even number (e.g., 6.25 rounds to 6.2).

□ Preceding number is increased by one if it is an odd number (e.g., 6.35 rounds to 6.4).

• Precision and Accuracy:

o Precision: Refers to the closeness of various measurements for the same quantity.

o Accuracy: The agreement of a particular value to the true value of the result.

o Example: True value 2.00 g. Student A (1.95 g, 1.93 g) - precise but not accurate. Student B (1.94 g, 2.05 g) - neither precise nor accurate. Student C (2.01 g, 1.99 g) - both precise and accurate.

• Dimensional Analysis (Factor Label Method / Unit Factor Method):

o Method used to convert units from one system to another.

o Involves multiplying by unit factors (ratios equal to 1, derived from equivalences like 1 in = 2.54 cm).

o Units can be handled like numerical parts (cancelled, divided, multiplied, squared).

10. Laws of Chemical Combinations

• These five basic laws govern the combination of elements to form compounds:

1. Law of Conservation of Mass:

□ Put forth by Antoine Lavoisier in 1789.

□ States that in all physical and chemical changes, matter can neither be created nor destroyed.

□ No net change in mass during the process.

□ Formed the basis for later developments in chemistry.

2. Law of Definite Proportions (Law of Definite Composition):

□ Given by French chemist Joseph Proust.

□ States that a given compound always contains exactly the

same proportion of elements by weight (or mass), regardless of its source.

□ Example: Cupric carbonate, whether natural or synthetic, always has 51.35% copper, 9.74% carbon, and 38.91% oxygen.

3. Law of Multiple Proportions:

□ Proposed by Dalton in 1803.

□ If two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in a ratio of small whole numbers.

□ Example: Hydrogen and oxygen form water (2g H with 16g O) and hydrogen peroxide (2g H with 32g O). The oxygen masses (16g and 32g) with fixed hydrogen (2g) are in a 1:2 ratio.

4. Gay-Lussac's Law of Gaseous Volumes:

□ Given by Gay Lussac in 1808.

□ When gases combine or are produced in a chemical reaction, they do so in a simple ratio by volume, provided all gases are at the same temperature and pressure.

□ Example: 100 mL hydrogen + 50 mL oxygen $\rightarrow$ 100 mL water vapor (volumes in 2:1 ratio).

□ This law is considered the law of definite proportions by volume and was later explained by Avogadro's work.

5. Avogadro's Law:

□ Proposed by Avogadro in 1811.

□ States that equal volumes of all gases at the same temperature and pressure should contain equal numbers of molecules.

□ Avogadro made a crucial distinction between atoms and molecules.

□ This law explained Gay-Lussac's results by considering molecules to be polyatomic (e.g., diatomic hydrogen and oxygen). Despite being correct, it initially did not gain much

support. Stanislao Cannizzaro emphasized its importance in 1860.

11. Dalton's Atomic Theory

- The idea of indivisible particles (a-tomio) dates back to Democritus (460–370 BC).
- In 1808, Dalton published 'A New System of Chemical Philosophy'.
- Postulates:
 1. Matter consists of indivisible atoms.
 2. All atoms of a given element have identical properties, including identical mass; atoms of different elements differ in mass.
 3. Compounds are formed when atoms of different elements combine in a fixed ratio.
 4. Chemical reactions involve the reorganization of atoms; atoms are neither created nor destroyed in a chemical reaction.
- Dalton's theory explained the laws of chemical combination but could not explain the laws of gaseous volumes or the reason for atoms combining.

12. Atomic and Molecular Masses

- Atoms are extremely small, so their masses are very low.
- Atomic Mass:
 - o Modern system is based on carbon-12 (^{12}C) as the standard, agreed upon in 1961.
 - o ^{12}C is assigned a mass of exactly 12 atomic mass unit (amu).
 - o Masses of all other atoms are given relative to this standard.
 - o One atomic mass unit (amu) is defined as a mass exactly equal to one-twelfth of the mass of one carbon-12 atom.
 - o $1 \text{ amu} = 1.66056 \times 10^{-24} \text{ g}$.
 - o 'amu' has been replaced by 'u' (unified mass).
- Average Atomic Mass:

- o Most naturally occurring elements exist as multiple isotopes.
- o Average atomic mass is computed by considering the existence of these isotopes and their relative abundance (percent occurrence).
- o Example: Carbon (^{12}C , ^{13}C , ^{14}C) has an average atomic mass of 12.011 u.
- o Atomic masses in the periodic table represent average atomic masses.
- Molecular Mass:
 - o The sum of atomic masses of all elements present in a molecule.
 - o Obtained by multiplying the atomic mass of each element by the number of its atoms and adding them.
 - o Example: Molecular mass of methane (CH_4) = (12.011 u) + 4(1.008 u) = 16.043 u.
 - o Molecular mass of water (H_2O) = 2(1.008 u) + 16.00 u = 18.02 u.
 - o Molecular mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) = 180.162 u.
- Formula Mass:
 - o Used for substances that do not contain discrete molecules as constituent units (e.g., ionic compounds like sodium chloride, NaCl).
 - o In such compounds, positive (Na^+) and negative (Cl^-) ions are arranged in a 3D structure.
 - o Calculated as the sum of atomic masses of the elements in the formula unit.
 - o Example: Formula mass of sodium chloride (NaCl) = atomic mass of Na + atomic mass of Cl = 23.0 u + 35.5 u = 58.5 u.

13. Mole Concept and Molar Masses

- Atoms and molecules are extremely small, so their numbers are very large.
- Mole (mol):

- o A unit of convenient magnitude to count entities at the microscopic level (atoms, molecules, particles, electrons, ions).
- o The SI unit for the amount of substance.
- o One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. This is the fixed numerical value of the Avogadro constant (N_a), also called the Avogadro number.
- o An elementary entity can be an atom, molecule, ion, electron, or any specified group of particles.
- o The mole of a substance always contains the same number of entities, regardless of the substance.
- o Example: 1 mol of hydrogen atoms = 6.022×10^{23} atoms; 1 mol of water molecules = 6.022×10^{23} water molecules.
- Molar Mass:
 - o The mass of one mole of a substance in grams.
 - o Numerically equal to its atomic, molecular, or formula mass in unified mass (u).
 - o Example: Molar mass of water = 18.02 g mol^{-1} ; Molar mass of sodium chloride = 58.5 g mol^{-1} .

14. Percentage Composition

- Provides information about the percentage of each element present in a compound.
- Useful for determining the formula of an unknown compound or checking the purity of a known compound.
- Calculation: Mass % of an element = $(\text{mass of that element in the compound} / \text{molar mass of the compound}) \times 100$.
- o Example: For water (H_2O , Molar Mass = 18.02 g):
 - Mass % of hydrogen = $(2 \times 1.008 \text{ g} / 18.02 \text{ g}) \times 100 = 11.18\%$
 - Mass % of oxygen = $(16.00 \text{ g} / 18.02 \text{ g}) \times 100 = 88.79\%$

15. Empirical Formula and Molecular Formula

- Empirical Formula: Represents the simplest whole number ratio of various atoms present in a compound.

- **Molecular Formula:** Shows the exact number of different types of atoms present in a molecule of a compound.
- **Determining from Percentage Composition:**
 1. Convert mass percent to grams (assuming 100 g sample).
 2. Convert grams of each element into moles by dividing by respective atomic masses.
 3. Divide each mole value by the smallest number of moles obtained to get a simple ratio.
 4. Write the empirical formula using these ratios.
 5. Determine empirical formula mass.
 6. Divide the given molar mass by the empirical formula mass to find 'n' (a whole number).
 7. Multiply the empirical formula by 'n' to get the molecular formula.

16. Stoichiometry and Stoichiometric Calculations

- **Stoichiometry:** Derived from Greek words meaning "element" and "measure".
- Deals with the calculation of masses (and sometimes volumes) of reactants and products involved in a chemical reaction.
- **Balanced Chemical Equation:**
 - o Has the same number of atoms of each element on both sides (Law of Conservation of Mass).
 - o Coefficients (stoichiometric coefficients) represent the number of molecules (and moles) taking part in or formed in the reaction.
 - o Provides information on:
 - Moles of reactants and products (e.g., $1 \text{ mol CH}_4 + 2 \text{ mol O}_2 \rightarrow 1 \text{ mol CO}_2 + 2 \text{ mol H}_2\text{O}$).
 - Number of molecules (e.g., $1 \text{ molecule CH}_4 + 2 \text{ molecules O}_2 \rightarrow 1 \text{ molecule CO}_2 + 2 \text{ molecules H}_2\text{O}$).
 - Volumes of gases at same T & P (e.g., $22.7 \text{ L CH}_4 + 45.4$

$L O_2 \rightarrow 22.7 L CO_2 + 45.4 L H_2O$).

□ Masses (e.g., $16 g CH_4 + 2 \times 32 g O_2 \rightarrow 44 g CO_2 + 2 \times 18 g H_2O$).

o States of matter are indicated by (s) for solid, (l) for liquid, (g) for gas, (aq) for aqueous.

o Subscripts in formulas cannot be changed to balance an equation.

• Limiting Reagent:

o In reactions where reactants are not in stoichiometric amounts, one reactant will be present in a lesser amount than required.

o This reactant gets consumed first and limits the amount of product formed.

o It is crucial to identify the limiting reagent for accurate stoichiometric calculations.

17. Reactions in Solutions (Concentration)

• Many laboratory reactions are carried out in solutions.

• Ways to express the amount of substance (concentration) in a solution:

1. Mass per cent (or weight per cent, w/w %):

□ Mass % of solute = $(\text{Mass of solute} / \text{Mass of solution}) \times 100$.

2. Mole Fraction (x):

□ Ratio of the number of moles of a particular component to the total number of moles of the solution.

□ For substance A dissolving in B: $x_a = n_a / (n_a + n_b)$.

3. Molarity (M):

□ Most widely used unit.

□ Defined as the number of moles of solute in 1 litre of the solution.

□ Molarity (M) = Moles of solute / Volume of solution (L).

□ Temperature-dependent because volume is temperature-

□ Dilution formula: $M_1V_1 = M_2V_2$ (where M and V are molarity and volume).

4. Molality (m):

□ Defined as the number of moles of solute present in 1 kg of solvent.

□ Molality (m) = Moles of solute / Mass of solvent (kg).

□ Does not change with temperature since mass is unaffected by temperature.

